MILESTONE 2

Deep Learning-Based Human Face Authenticity Detection

Comprehensive Data Preparation, EDA & Preprocessing

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Dual-Stream Vision Transformer | GAN Deepfake Detection

Objectives for Milestone 2

OVERVIEW

01

Dataset Sourcing & Documentation

Formally define the dataset, its origins, licenses, and inherent characteristics.

02

Exploratory Data Analysis (EDA)

Statistical & visual analysis — pixel distributions, aspect ratios, class balance.

03

Data Preprocessing

Standardise images for CNN/ViT ingestion: resize to 224×224, pixel normalisation.

04

Data Augmentation

Simulate real-world degradation: compression, lighting shifts, geometric distortions.

05

Train / Val / Test Split

Finalise rigidly isolated data partitions in a Keras-compliant directory structure.

Outcome

Data pipeline 100% prepared for Milestone 3 model experimentation & architecture benchmarking.

Dataset Documentation

DATASET

Dataset Name	Real vs AI Generated Faces Dataset
Contributor	Kaggle — philosopher0808
Total Images	120,000+ high-resolution facial images
Class 0 – Real	Authentic photographs (FFHQ — Flickr-Faces-HQ)
Class 1 – Fake	Synthetic faces via StyleGAN / StyleGAN2
License	CC BY-NC-SA 4.0 · Academic / research use

Why This Dataset?

FFHQ Authentic Faces: Extreme variance in age, ethnicity, lighting & pose — forces model to learn biological structure, not shortcuts.

StyleGAN2 Synthetic Faces: Fool even expert human reviewers — no blurry ears or mismatched reflections that plagued early GANs.

Perfectly Balanced Classes: No resampling (SMOTE) or weighted loss needed — Standard Cross-Entropy applies directly.

Pre-cropped & Aligned: All images are perfect squares — resize to 224×224 without padding or distortion.

EDA: Class Distribution & Balance

EDA

~50% Real Faces (FFHQ)

~50% AI-Generated (StyleGAN2)

120K+ Total Dataset Size

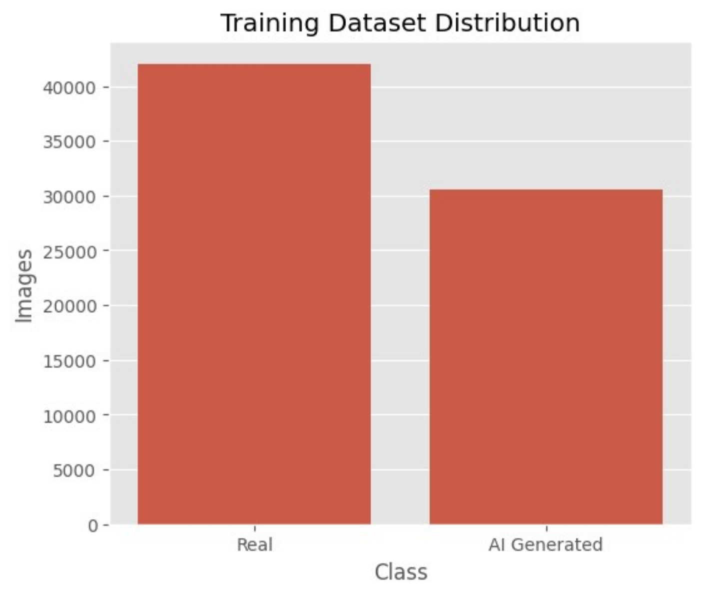

Fig 1: Balanced Training Dataset Distribution

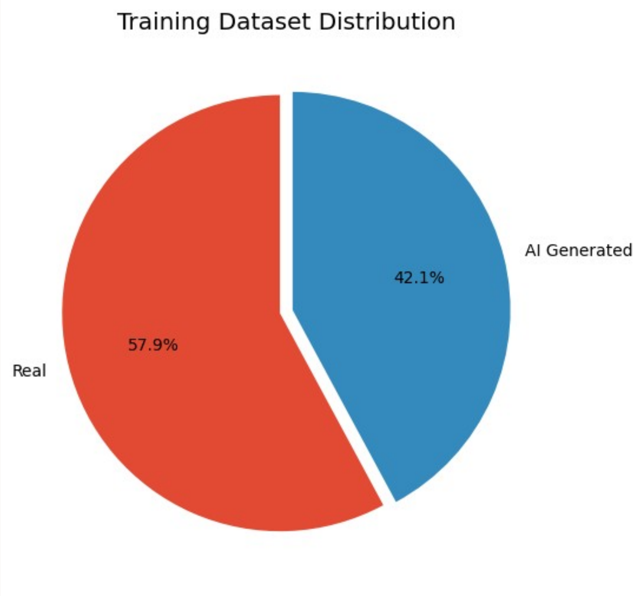

Fig 2: Confirmed Proportional Split (Real vs AI)

EDA: Visual Inspection & Dataset Complexity

EDA

The Forgery Challenge

— AI-generated faces exhibit extraordinary realism. Skin texture, lighting physics, and facial geometry are mathematically precise.

— Early synthetic artifacts (like blurry ears or asymmetric eyes) are completely bypassed by StyleGAN2 style-based generators.

— Standard edge detectors or HOG features will fail entirely against these modern generative methods.

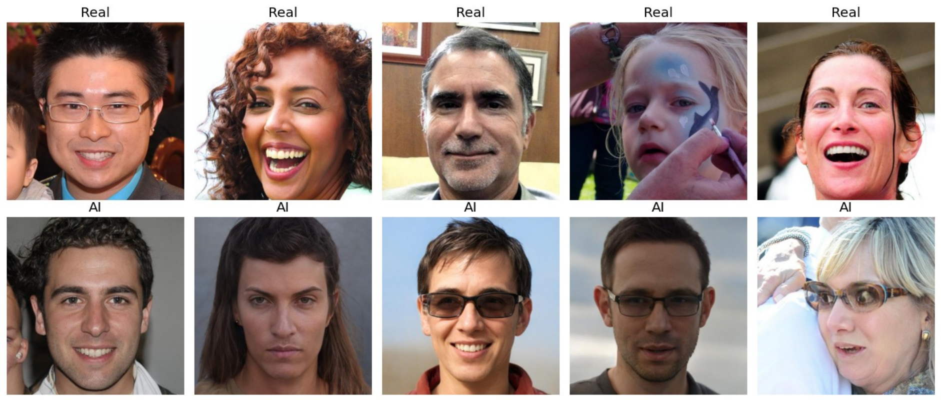

Fig 3: Random Sample Grid — Real Faces (Top) vs AI-Generated (Bottom)

CONCLUSION: Modern deep learning classification (dual-stream CNNs or Vision Transformers) is required to detect high-dimensional forensic clues.

EDA: Aspect Ratio & Pixel Intensity Analysis

Aspect Ratio Distribution

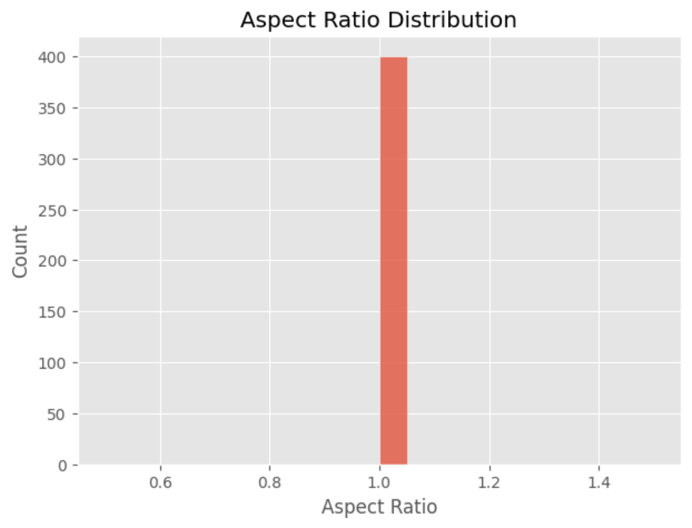

Fig 4: Near-perfect 1:1 aspect ratio across 400 sampled images.

All images are perfect squares → safe to resize directly to 224×224 with no distortion or padding.

RGB Pixel Intensity Histogram

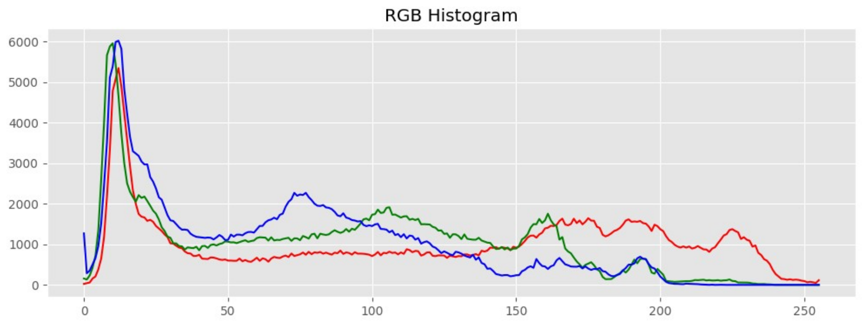

Fig 5: Full dynamic range (0–255) across R, G, B channels.

- ✓ Full 8-bit dynamic range utilised — rich spectral detail preserved
- ✓ Validates pixel normalisation (÷ 255) without range compression
- ✓ No clipping or saturation artifacts in source imagery

Image Preprocessing Pipeline

PREPROCESSING

01

Pixel Normalisation

```
rescale = 1./255
```

Every pixel squashed to [0.0–1.0] float32. Stabilises gradient descent, prevents exploding/vanishing gradients, and keeps activations (ReLU / GELU) in their stable range.

02

Spatial Standardisation

```
target_size = (224,224)
```

224×224 is the de-facto ImageNet standard. Guarantees hot-swap compatibility across ResNet, EfficientNet-B4, ViT-B/16 — no pipeline rebuild in M3.

03

Batching Strategy

```
batch_size = 32
```

Optimal GPU VRAM balance + SGD stability. Smaller batches → erratic loss curves. Very large batches → sharp local minima. 32 is the proven sweet spot.

Fig 6: 9-image subset from a (32, 224, 224, 3) preprocessed batch tensor.

Robust Data Augmentation Strategy

AUGMENTATION

rotation_range = 20°	Simulates non-level cameras & tilted head postures
width/height_shift = 15%	Achieves spatial invariance — model scans full frame equally
shear_range = 15%	Mimics wide-angle lens distortion & angular offsets
zoom _range = 20%	Robustness across crops, close-ups & focal lengths
horizontal_flip = True	Doubles variation; disrupts GAN noise memorisation
brightness_range [0.8–1.2]	Prevents model correlating brightness level with class

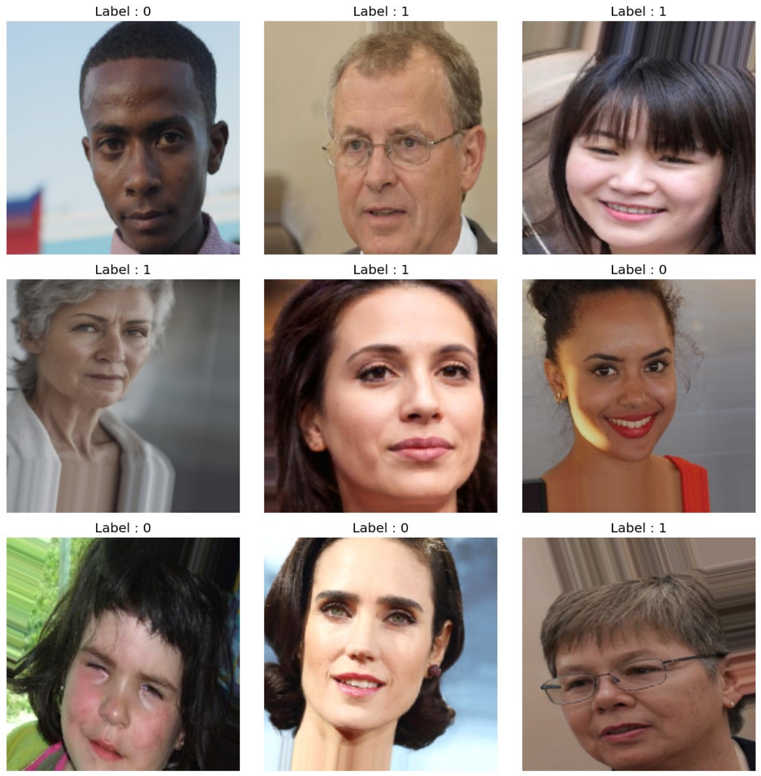

Fig 7: Augmented Training Dataset Variety

Augmentation: Original vs Augmented

The combination of rotation, zoom, and brightness adjustments drastically alters the underlying pixel arrangement while preserving the core biological features of the human face.

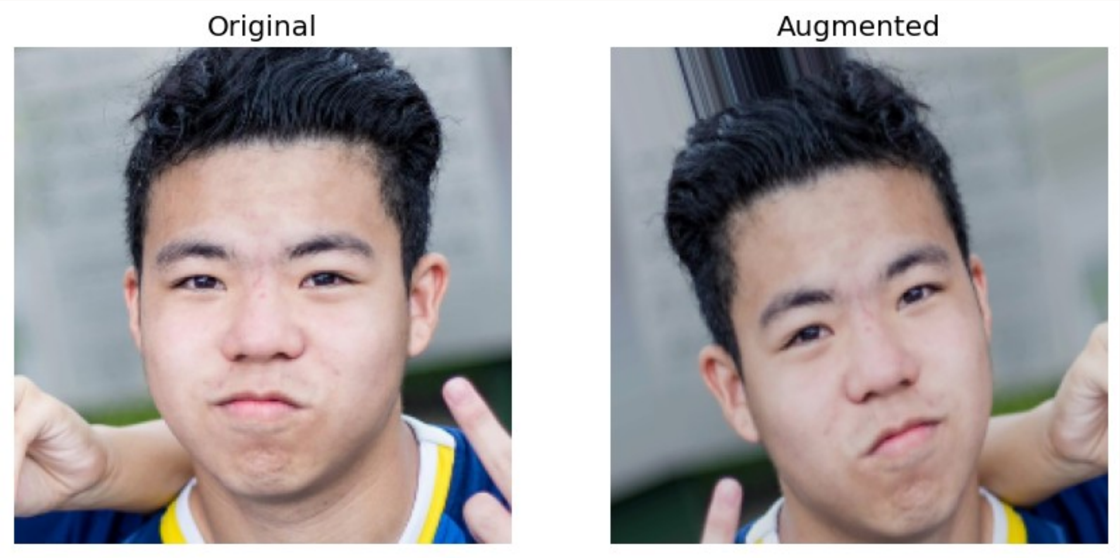

Fig 8: Direct Side-by-Side Comparison of Original Image (Left) and its Augmented Counterpart (Right)

CRITICAL ISOLATION MANDATE: Validation and testing datasets must never be augmented. Only rescale=1./255 is applied. Augmentation forces robust feature learning during training, but validation/testing must measure performance on clean, unaltered real-world distributions.

Train / Validation / Test Split & Folder Structure

DATA SPLIT

TRAIN

~70%

Backpropagation & weight updates

VALIDATE

~15%

Epoch-end hyperparameter tuning

Generator Verification

TEST

```
/dataset/
├── train/      ← Backpropagation & weight updates
│   ├── 0/      ← Real Faces
│   └── 1/      ← AI Generated
├── validate/   ← Epoch-end tuning
│   ├── 0/
│   └── 1/
└── test/       ← Held-out final benchmarking
    ├── 0/
    └── 1/
```

- ✓ Training samples accurately loaded & mapped to classes 0 and 1
- ✓ Validation set strictly isolated — zero overlap with train
- ✓ Test set held completely isolated until M3 benchmarking
- ✓ Batch step count auto-computed: samples ÷ 32 per epoch
- ✓ Data leakage verified absent — scientifically valid splits

Source: kaggle.com/datasets/philosopher0808/real-vs-ai-generated-faces-dataset

Conclusion & Readiness for Milestone 3

CONCLUSION

Dataset Documented & Verified

Rigorously profiled — perfectly balanced, high-variance FFHQ + StyleGAN2 sourced. Licensing confirmed for academic use.

Preprocessing Pipeline Built

224×224 spatial standardisation + pixel normalisation to [0.0, 1.0] — absolute prerequisite for stable gradient descent.

Defensive Augmentation Applied

Dynamic rotation, shifts, zoom, flip & brightness adjustments prevent overfitting and enforce generalisation to real-world inputs.

Strict Data Isolation Enforced

Zero data leakage across Train / Val / Test partitions. Evaluation metrics will be scientifically valid and reproducible.

Data is 100% ready — streaming normalised (32, 224, 224, 3) tensors directly into the Dual-Stream ViT & CNN architectures in Milestone 3.

Team Declaration

M2 · 2026

We certify that all team members have actively contributed to the preparation of this milestone report. Each member has reviewed the contents, understands the work presented, and agrees with the submitted report.

R	R	V	A	S
Rohit	**Raunak**	**Vishakha**	**Aman**	**Somendu**
Reviewed & Signed	Reviewed & Signed	Reviewed & Signed	Reviewed & Signed	Reviewed & Signed
✓	✓	✓	✓	✓